Xinghao Yao

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Research Fields Primary: Behavioural Economics; AI Economics.
Secondary: Experimental Economics; Information Economics; Auctions and Market Design.

Education

The University of Queensland **2023-present**
PhD in Economics (expected early 2027)
Supervisors: Kenan Kalayci and Lionel Page.

The University of Queensland **2022**
Master of Advanced Economics

The University of Queensland **2018-2021**
Master of Applied Econometrics

Job Market Paper

How Robust Is LLM Economic Decision-Making? An Auction Benchmark
with Kenan Kalayci

We examine the robustness of economic decision-making by large language models (LLMs) using a theory-grounded benchmark of 78 auction environments spanning bidding, participation, and information purchase. Our findings reveal substantial variation in decision accuracy across auction environments and demonstrate that performance in one decision type does not necessarily generalize to others. This dependence on the decision environment extends to how LLMs respond to additional information that leaves theoretical predictions unchanged. It also qualifies the benefits of reasoning: enabling reasoning improves overall performance in the tested comparisons, but its gains remain uneven across tasks. These results support evaluating LLMs for decision-making against the conditions and connected decisions relevant to their intended use, rather than inferring general competence from isolated successes or aggregate scores.

Working Papers

Information Disclosure in Private-Value Auctions
with Kenan Kalayci | *Working paper*

Information shapes what people know, but outcomes depend on what they do. In auctions, this distinction matters: an object goes to the highest bidder, not necessarily to the person who values it most. We experimentally study how information disclosure affects bidding and allocative efficiency in private-value second-price auctions. A bidder receives partial information about her private value and reports incentivized value estimates before bidding. We compare random disclosure, directionally selected disclosure, and full value revelation between estimation and bidding. Directional disclosure produces more efficient allocations and lower welfare losses than random disclosure with three and six bidders. Participants' estimates also rank private values more accurately under directional than random disclosure. With six bidders, directional disclosure yields lower welfare losses than full revelation, with fewer inefficient wins by low-value bidders and smaller losses from these wins. Full revelation removes uncertainty about value, but not departures from truthful bidding. The allocation benefits of disclosure depend on how information shapes the bids that determine the winner.

Working Papers

What You See Is All There Is? A Registered Direct Replication of Enke (2020)
R&R | Journal of the Economic Science Association

This paper reports a registered direct replication of Enke (2020) in a different language, laboratory and software implementation. Participants estimate the mean of a full set of signals after observing a selected subset. Those who must infer the implications of hidden signals place excessive weight on the mean of visible signals compared with participants given an explicit summary of the hidden information. The replication supports the main finding, but provides only limited support for the proposed mechanism that unobserved information is neglected because it fails to come to mind. Greater computational complexity increases such neglect.

Sponsorship Disclosure as a Strategic Informational Cue: A Two-Sided Labelling Model and Meta-Analysis of Influencer Marketing
with Tianyuan Feng | *Major revision | Journal of the Academy of Marketing Science (FT50)*

This paper integrates a two-sided labelling model with a three-level meta-analysis of 63 articles, 89 studies and 258 effect sizes. It studies how sponsorship disclosure affects consumer responses and how these effects vary across products, influencers and audiences. The evidence points to persuasion knowledge as a central mechanism behind the negative average effect of disclosure on consumer responses, relative to no disclosure. My contribution was to develop the paper's two-sided labelling model.

From Analysis to Affect: Modeling and Measuring Finfluencer Communication with Generative LLMs
with Tianyuan Feng | *Under review | Journal of the Academy of Marketing Science (FT50)*

Research on financial influencers (finfluencers) largely measures the tone of their posts, not how the posts argue: with analysis or through emotion. We use a codebook-guided large language model to classify how 47,707 posts by 396 finfluencers on X argue. Analytical-only posts are most common, yet posts expressing emotion draw more likes and replies. Where emotion dominates, this advantage is three to four times greater among large finfluencers, a gap that persists after controlling for views. We model finfluencers' choice between analysis and emotion: emotional posts are easier to act on, but analysis builds a reputation for competence that lifts engagement with every post, explaining why they keep posting analysis and how a large following can widen emotion's advantage. Experiments trace liking to perceived arousal and following to perceived competence. Because platforms favor engagement, the content reaching retail investors may tilt from analysis toward affect, especially from large finfluencers.

My contribution comprises the economic modelling and generative LLM-based textual analysis, including the classification of semantic content and discrete emotion frames.

Work in Progress

Accuracy or Affirmation? Competition in AI Advice Markets

Underquoting in Endogenous Entry Auctions

Do you know how attractive you are? A Two-Sided Search Model with Self-Uncertainty

Research Collaboration	**Project Silicon** Contribute experimental data and replication materials, and review simulation results based on our experiments.	
Teaching Experience	**Tutor, The University of Queensland** Microeconomic Analysis: student evaluation 4.93/5. Introduction to Econometrics: student evaluation 4.8/5.	**2023-2026**
Conferences	Asia-Pacific ESA Meeting, University of Melbourne | Presenter	2026
	Australasian Economic Theory Workshop (AETW), Auckland | Presenter	2026
	Australian PhD Conference in Economics and Business, UNSW | Presenter	2025
	Summer School in Experimental and Behavioral Economics, University of Crete | Participant	2025
	Foundations of Utility and Risk (FUR), The University of Queensland | Presenter	2024
Other Related Experience	Research Assistant, UQ QSMAT program.	
	Summer Research Scholar, The University of Queensland	2019-2020
	Master's thesis: constructed a Chinese financial-text corpus incorporating domain-specific expressions for sentiment classification and predicting the direction of stock price movements.	
	DiDi, Safety Department: behavioural analysis	Jun-Nov 2017
Research Skills and Languages	Methods: laboratory experiments, applied econometrics, game-theoretic modelling, machine learning and natural language processing.	
	Software: Python (proficient); Stata, R and MATLAB (working knowledge); oTree.	
	Languages: Mandarin (native); English (fluent).	